ERA: Evidence-based Reliability Alignment for Honest Retrieval-Augmented Generation

Sunguk Shin
0000-0002-3424-405X
Korea University , Seoul , South Korea
MPI-SP , Bochum , Germany
ssw1419@korea.ac.kr

Meeyoung Cha
0000-0003-4085-9648
MPI-SP , Bochum , Germany
KAIST , Daejeon , South Korea
mia.cha@mpi-sp.org

Byung-Jun Lee
Co-corresponding authors 0000-0002-0684-607X
Korea University , Seoul , South Korea
Gauss Labs Inc. , Seoul , South Korea
byungjunlee@korea.ac.kr

Sungwon Park
0000-0002-6369-8130
KAIST , Daejeon , South Korea
MPI-SP , Bochum , Germany
psw0416@kaist.ac.kr

2026; © acmlicensed

Abstract.

Retrieval-Augmented Generation (RAG) grounds language models in factual evidence but introduces critical challenges regarding knowledge conflicts between internalized parameters and retrieved information. However, existing reliability methods, typically relying on scalar confidence, fail to explicitly distinguish between epistemic uncertainty and inherent data ambiguity in such hybrid scenarios. In this paper, we propose a new framework called ERA (Evidence-based Reliability Alignment) to enhance abstention behavior in RAG systems by shifting confidence estimation from scalar probabilities to explicit evidence distributions. Our method consists of two main components: (1) Contextual Evidence Quantification, which models internal and external knowledge as independent belief masses via the Dirichlet distribution, and (2) Quantifying Knowledge Conflict, which leverages Dempster-Shafer Theory (DST) to rigorously measure the geometric discordance between information sources. These components are used to disentangle epistemic uncertainty from aleatoric uncertainty and modulate the optimization objective based on detected conflicts. Experiments on standard benchmarks and a curated generalization dataset demonstrate that our approach significantly outperforms baselines, optimizing the trade-off between answer coverage and abstention with superior calibration.

1. Introduction

Recent advances in Large Language Models (LLMs), trained on massive corpora, have profoundly transformed the landscape of Natural Language Processing (NLP). Despite their remarkable capabilities, these models remain prone to hallucinations, stemming from their reliance on static training data that may become outdated and their tendency to generate factually incorrect or misleading content (Kalai et al. 2025; Zhang et al. 2024). Retrieval-Augmented Generation (RAG) has emerged as a promising solution by incorporating external knowledge sources into the generation process, thereby grounding model outputs in factual evidence and improving overall reliability (Lewis et al. 2020).

However, despite its advantages, RAG introduces a new class of challenges arising from its hybrid nature. In particular, conflicts may emerge between a model's internalized parametric knowledge and the retrieved external information (Longpre et al. 2021; Mallen et al. 2023). Such discrepancies can create ambiguity during generation, making it difficult for the model to reconcile conflicting evidence. As a result, the model may exhibit inconsistent behavior, degraded response quality, or erroneous reasoning when it fails to appropriately prioritize the correct information source (Shin and Kim 2025). Addressing this challenge is critical for ensuring the trustworthiness of retrieval-augmented systems in real-world applications.

To mitigate these issues, prior work has explored mechanisms for regulating a model's reliance on retrieved information, including dynamic gating strategies. In parallel, early research investigated unsupervised approaches for assessing response reliability through statistical proxies. For example, Guerreiro et al. leveraged token-level output probabilities—such as average or minimum logit values—to estimate generation confidence (Guerreiro et al. 2023), while Wang et al. proposed Self-Consistency, which samples multiple reasoning paths and selects the most reliable answer via majority voting (Wang et al. 2023). In contrast, another line of work adopts a supervised paradigm. For instance, Sun et al. employed Direct Preference Optimization (DPO) to train models to abstain from answering when retrieved context is insufficient, using curated abstention datasets to align model behavior with reliability objectives (Sun et al. 2025).

Despite these advances, existing approaches remain vulnerable to overfitting—either to the pre-training distribution in unsupervised methods or to specific abstention patterns in supervised ones. As a result, unsupervised methods often struggle to distinguish confident hallucinations from correct predictions, while supervised approaches may memorize refusal behaviors, limiting their generalization to unseen scenarios. These shortcomings stem from two fundamental challenges. First, conventional probabilistic frameworks rely on scalar confidence estimates that fail to distinguish epistemic uncertainty (arising from lack of knowledge) from aleatoric uncertainty (stemming from inherent data ambiguity), often leading to overconfident predictions (Guo et al. 2017; Kuhn et al. 2023; Park et al. 2025). Second, the aggregation of internal model knowledge and external retrieved contexts into a single output distribution obscures conflicts between these information sources, hindering the reliable identification of hallucinations (Mallen et al. 2023; Jin et al. 2025). This highlights the critical need for a robust quantification metric capable of explicitly disentangling belief from uncertainty while simultaneously measuring the conflict between internal parametric knowledge and external retrieved evidence.

To address these limitations, we propose ERA (Evidence-based Reliability Alignment), a principled framework designed to enhance abstention behavior in RAG systems. Inspired by evidential deep learning (Sensoy et al. 2018), ERA shifts confidence estimation from scalar probabilities to explicit modeling of evidence distributions. By leveraging the Dirichlet distribution to parameterize evidence density, our approach disentangles epistemic and aleatoric uncertainty in a principled manner. Furthermore, ERA independently models belief masses derived from parametric knowledge and retrieved evidence. Using Dempster–Shafer Theory (DST) (Jsang 2018), it quantitatively measures belief conflict—the disagreement between internal priors and external information—thereby providing a rigorous geometric foundation for reliable and interpretable abstention.

Experiments demonstrate that ERA optimizes the trade-off between answer coverage and abstention, maintaining high accuracy on answerable queries while selectively withholding responses in uncertain or conflicting scenarios. In comparisons with several recent baselines on standard RAG benchmarks, our results show that the model significantly improves response reliability without compromising its ability to answer supported queries. Furthermore, this robustness is validated through evaluations on a curated dataset of Wikipedia events, specifically designed to assess generalizability. Finally, extensive empirical analysis confirms that ERA produces well-calibrated confidence estimates, ensuring that its predicted uncertainty accurately reflects the true likelihood of correctness.

The major contributions and findings of our work include:

- Proposing a novel reliability alignment framework, ERA, that fundamentally shifts confidence quantification from scalar probabilities to evidence distributions via the Dirichlet distribution, explicitly disentangling epistemic uncertainty from aleatoric uncertainty.
- Introducing a rigorous quantification of 'belief conflict' by leveraging Dempster-Shafer Theory (DST) to measure the geometric discordance between the model's internal parametric knowledge and external retrieved evidence, which are modeled as independent belief masses.
- Constructing a curated benchmark dataset derived from Wikipedia events specifically designed to evaluate generalization, verifying that ERA robustly identifies unsupported queries across diverse domains without overfitting to specific refusal patterns.
- Demonstrating through extensive experiments that ERA produces well-calibrated confidence estimates and optimizes the trade-off between answer coverage and abstention, significantly outperforming existing baselines in enabling honest RAG.

The training details and datasets are publicly available at `https://anonymous.4open.science/r/ERA-318B`.

2. Related work

2.1. Uncertainty Quantification and Evidential Deep Learning

In deep learning classifiers, softmax probabilities are widely used as a proxy for confidence; however, they are often miscalibrated and can be overconfident (Guo et al. 2017). To mitigate this miscalibration, various approaches have been proposed, ranging from approximate Bayesian inference methods (Gal and Ghahramani 2016) to Deep Ensembles (Lakshminarayanan et al. 2017; Park et al. 2021). While these methods often increase training cost due to sampling or multiple models, this added computational overhead can be undesirable for real-time or large-scale applications. To address this limitation, Evidential Deep Learning (EDL) predicts class-wise evidence and uses it to parameterize a Dirichlet distribution over class probabilities (Sensoy et al. 2018), enabling the model to compute both expected predictive probabilities and a corresponding measure of predictive uncertainty. This evidential approach has been further advanced by methods like Posterior Networks, which leverage latent-space normalizing-flow density estimation to improve the quality of evidence quantification (Charpentier et al. 2020).

Rather than treating class probabilities as a point estimate, EDL represents predictions as a Dirichlet distribution over class probabilities, thereby modeling uncertainty at the distribution level (Sensoy et al. 2018). Concretely, the network outputs non-negative class-wise evidence $e_k \geq 0$, which is mapped to Dirichlet concentration parameters via $\alpha_k = e_k + 1$. This formulation is closely related to Subjective Logic, where an opinion is decomposed into belief masses $\{b_k\}$ and an uncertainty mass u, satisfying the conservation constraint $u + \sum_k b_k = 1$ (Jsang 2018). Recent studies in Trusted Multi-View Classification demonstrate that fusing such evidential opinions from disparate sources significantly enhances reliability (Han et al. 2022). By separating supported belief from residual uncertainty, the evidential parameterization provides a structured representation of "knowledge" versus "ignorance".

To make the ignorance mass meaningful, EDL typically couples a data-fitting term with a regularization term that discourages unwarranted evidence, often implemented via a KL-divergence penalty toward a high-uncertainty reference Dirichlet distribution (Sensoy et al. 2018). This regularization encourages flatter predictive distributions when evidence is weak or ambiguous, reducing unsupported overconfidence and making evidential outputs more suitable for reliability-aware decision rules (Sensoy et al. 2021).

2.2. Honest RAG and Knowledge Conflicts

Hallucinations persist even in advanced language models and are not eliminated simply by augmenting generation with external tools like search or retrieval (Kalai et al. 2025; Lewis et al. 2020; Asai et al. 2024). Recent analyses argue that hallucinations stem from statistical pressures during pretraining (Kalai et al. 2025). These tendencies persist post-training because standard optimization rewards guessing over acknowledging uncertainty. Due to the tendency of LLMs to hallucinate (Kadavath et al. 2022; Zhang et al. 2024), the problem is merely mitigated when retrieval is accurate, but persists whenever the retrieved context is imperfect or absent. These observations motivate *honest RAG*, a paradigm focused on determining when retrieval-augmented systems should answer versus abstain (Lin et al. 2022), and how to prevent responses that are unsupported by evidence (Chen et al. 2024; Wu et al. 2024).

Within the scope of honest RAG, initial efforts have focused on measuring faithfulness violations. For instance, RAGTruth provides a corpus annotating cases where outputs conflict with or are unsupported by retrieved evidence (Niu et al. 2024; Maynez et al. 2020). This perspective shifts the diagnosis of RAG failures from "missing retrieval" to "unused or misused retrieval" (Jiang et al. 2023). Complementing this measurement-focused view, inference-time decoding interventions suggest adjusting decoding to reduce hallucinations (Chuang et al. 2024).

A parallel thread frames honest RAG as a risk-control or alignment problem. RC-RAG advocates for controlling the risk of answering under low confidence, analyzing factors like retrieval quality to motivate selective abstention (Chen et al. 2024). Similarly, Divide-Then-Align (DTA) formalizes the problem by defining boundaries between parametric knowledge and retrieval-provided knowledge (Sun et al. 2025). By partitioning data into quadrants (e.g., Known/Unknown Parametric vs. Known/Unknown Context), DTA aligns models to answer when supported and abstain when necessary.

However, a critical gap remains. Existing approaches (Chen et al. 2024; Sun et al. 2025) often treat "uncertainty" and "conflict" as outcomes to be detected or conditions for gating, rather than making them explicit, decomposed training signals. Specifically, they fail to explicitly distinguish between uncertain retrieval and strong retrieval that conflicts with strong parametric beliefs. Inference-time monitoring does not necessarily reshape the underlying policy, and risk-control methods may be conservative in some cases—potentially abstaining even when retrieval evidence is strong, e.g., when internal beliefs and retrieved evidence disagree (Mallen et al. 2023; Wen et al. 2024; Jin et al. 2025).

Our approach addresses this limitation by (i) separately estimating evidential support from parametric-only versus retrieval-augmented inputs, and (ii) converting their discrepancy into an explicit conflict signal. Rather than merely exposing the model to high-conflict cases during training, we use this explicit signal as a decomposed training signal to modulate alignment pressure, explicitly targeting disagreements between internal parametric beliefs and external evidence. More broadly, our method complements inference-time alignment or decoding-time interventions by shaping the training-time decision boundary for context utilization and abstention (Hadji-Kyriacou and Arandjelović 2024). This allows the model to effectively resolve conflicts by prioritizing reliable retrieval over misaligned priors. Consequently, our method operationalizes the critique that current incentives reward guessing, establishing a mechanism where abstention and context utilization are conditional on decomposed evidence and conflict resolution, rather than a single opaque confidence score.

3. PROBLEM FORMULATION

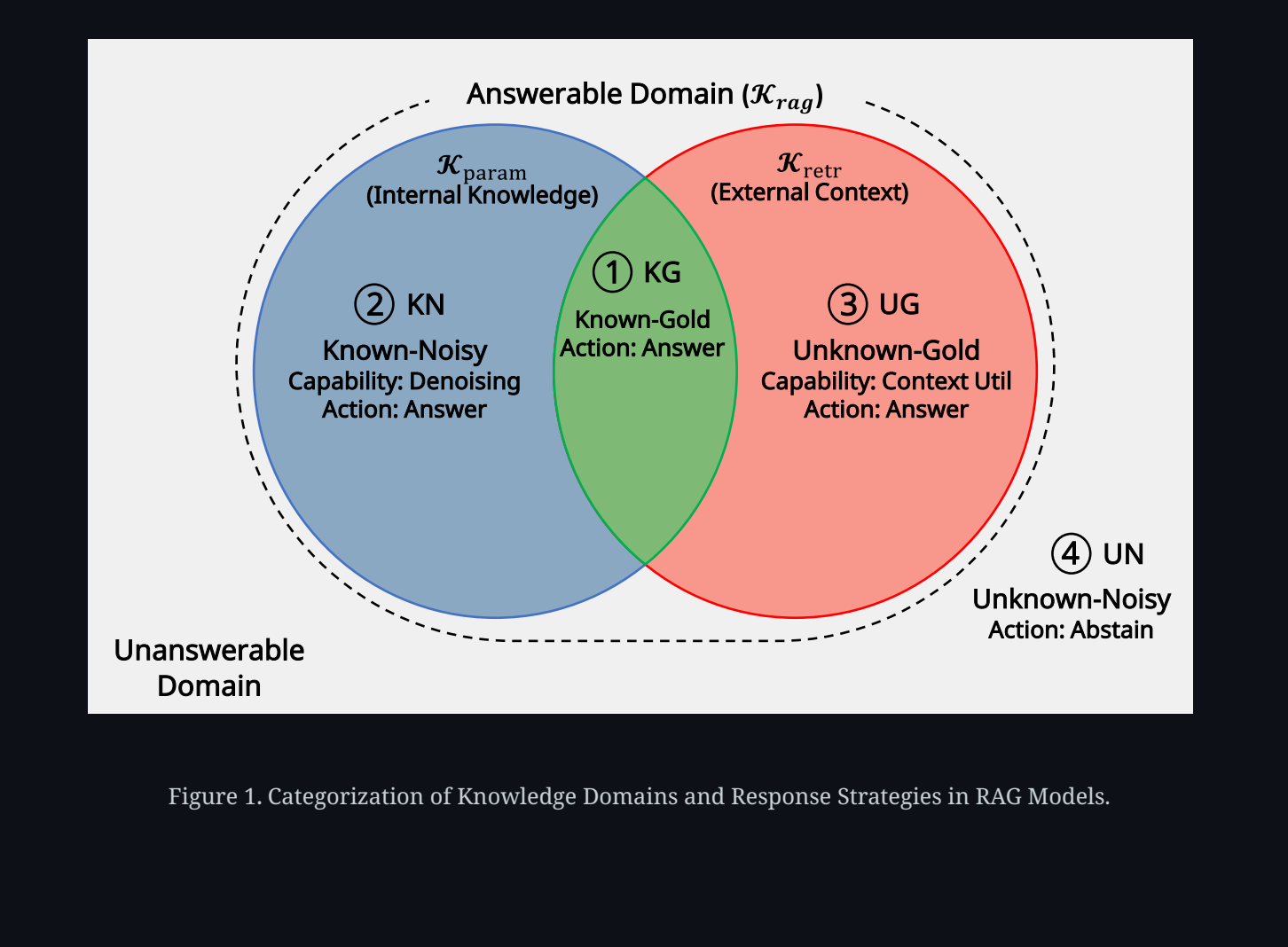


Figure 1. Categorization of Knowledge Domains and Response Strategies in RAG Models.

We formalize the problem of honest assessment, explicitly defining the knowledge boundaries, categorization criteria, and evaluation metrics employed in this study.

3.1. Honest Assessment

To rigorously evaluate the reliability of RAG systems, we first formalize the standard pipeline comprising a retriever $\mathcal{R}: \mathcal{Q} \to \mathcal{P}(\mathcal{D}_{\text{corpus}})$, which fetches relevant contexts $\mathcal{C} = \mathcal{R}(q) \subset \mathcal{D}_{\text{corpus}}$, and a generator $\mathcal{G}_\theta$. Let $\mathcal{Q}$ denote the query space and $\mathcal{A}$ the answer space. We define the generator $\mathcal{G}: \mathcal{Q} \times \mathcal{C} \to \mathcal{A}$ as a function producing an answer y by modeling the conditional probability $P(y|q, \mathcal{C})$. Additionally, let $g: \mathcal{Q} \to \mathcal{A}$ be the ground truth function mapping a query to its correct response, $C(\cdot): \mathcal{Q} \times \mathcal{A} \to \{0,1\}$ be a correctness evaluation function and $CT(\cdot): \mathcal{C} \times \mathcal{A} \to \{0,1\}$ be a context-containment indicator that returns 1 if the context c contains the ground-truth answer $g(q)$, and 0 otherwise.

Table 1. Summary of evaluation metrics.

Metric	Formula	Description
Answer Quality		
Recall	$\frac{\lvert\mathcal{S}_{\text{ans}} \cap \mathcal{T}_{\text{ans}}\rvert}{\lvert\mathcal{T}_{\text{ans}}\rvert}$	Correct answers over all answerable queries
Precision	$\frac{\lvert\mathcal{S}_{\text{ans}} \cap \mathcal{T}_{\text{ans}}\rvert}{\lvert\mathcal{S}_{\text{ans}}\rvert}$	Correct answers over all attempted answers
F1	$2 \cdot \frac{\text{Pre} \cdot \text{Rec}}{\text{Pre} + \text{Rec}}$	Harmonic mean of Precision and Recall
Retrieval Handling		
Denoise Rate	$\frac{\lvert\mathcal{S}_{\text{ans}} \cap \mathcal{T}_{\text{noise}}\rvert}{\lvert\mathcal{T}_{\text{noise}}\rvert}$	Accuracy on queries with noisy context
Context Util.	$\frac{\lvert\mathcal{S}_{\text{ans}} \cap \mathcal{T}_{\text{gold}}\rvert}{\lvert\mathcal{T}_{\text{gold}}\rvert}$	Accuracy on queries with golden context
Abstain Quality		
Abstain Rec.	$\frac{\lvert\mathcal{S}_{\text{abs}} \cap \mathcal{T}_{\text{unans}}\rvert}{\lvert\mathcal{T}_{\text{unans}}\rvert}$	Correct abstentions over unanswerable queries
Abstain Pre.	$\frac{\lvert\mathcal{S}_{\text{abs}} \cap \mathcal{T}_{\text{unans}}\rvert}{\lvert\mathcal{S}_{\text{abs}}\rvert}$	Correct abstentions over all abstained queries
Abstain F1	$2 \cdot \frac{\text{Pre} \cdot \text{Rec}}{\text{Pre} + \text{Rec}}$	Harmonic mean of Abstain Precision and Recall
Overall F1	$2 \cdot \frac{\text{F1} \cdot \text{F1}_{\text{abs}}}{\text{F1} + \text{F1}_{\text{abs}}}$	Trade-off between Answer and Abstain Quality

Knowledge Components. We posit that the system's total knowledge capability, denoted as $\mathcal{K}_{\text{rag}}$, is composed of two fundamental components: the parametric knowledge boundary of the LLM ($\mathcal{K}_{\text{param}}$) and the retrieval knowledge boundary ($\mathcal{K}_{\text{retr}}$). Formally, these are defined as:

$$\mathcal{K}_{\text{param}} = \{q \in \mathcal{Q} | C(\mathcal{G}(q, \varnothing)) = 1\} \quad \text{(Eq. 1)}$$

$$\mathcal{K}_{\text{retr}} = \{q \in \mathcal{Q} | \ \exists c \in \mathcal{R}(q) : CT(c, g(q)) = 1\} \quad \text{(Eq. 2)}$$

Consequently, the overall boundary is characterized as the union: $\mathcal{K}_{\text{rag}} = \mathcal{K}_{\text{param}} \cup \mathcal{K}_{\text{retr}}$. This formulation captures that a query is answerable if valid information resides within either the model's internal parameters or the retrieved external documents.

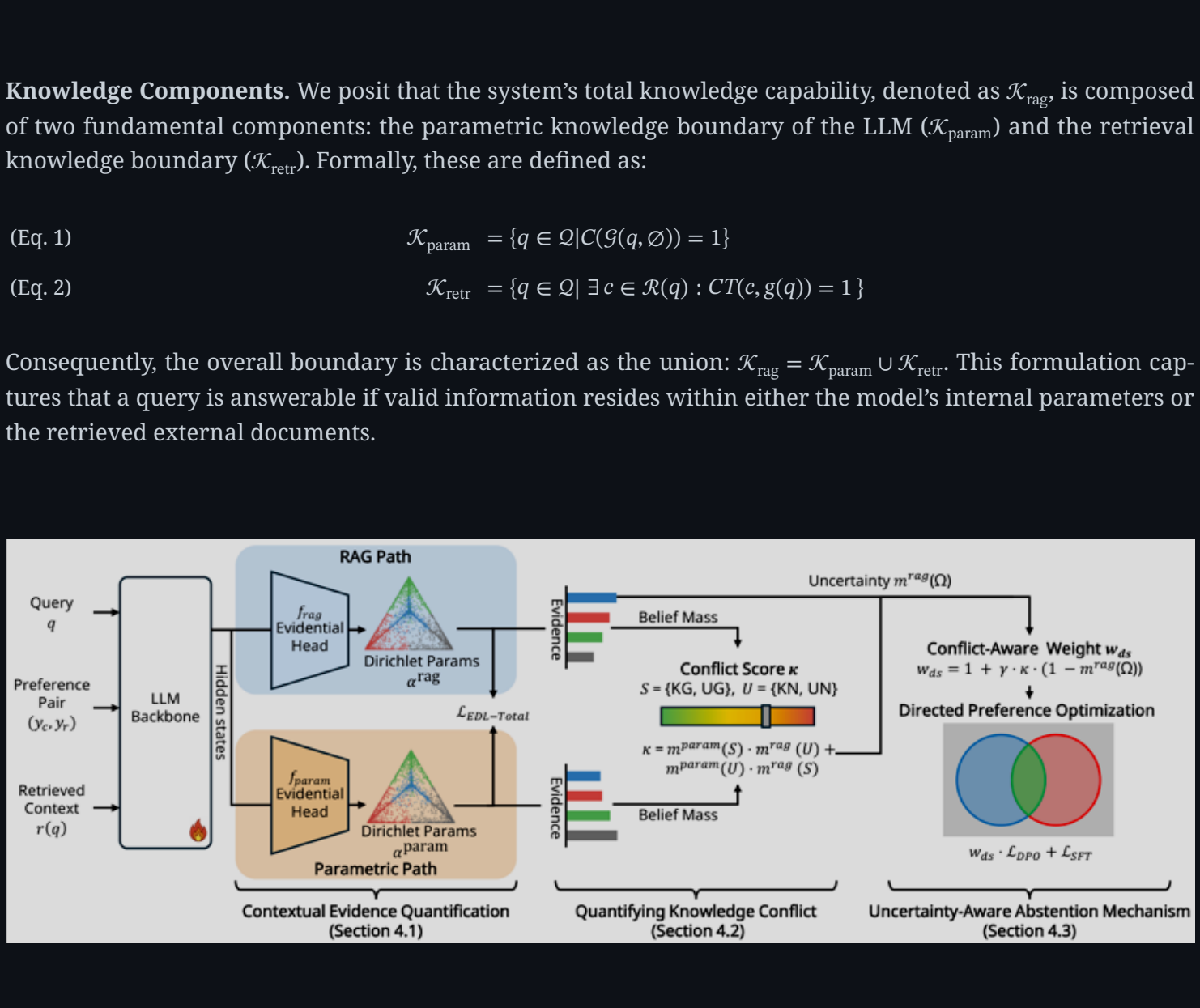

Query q
Preference Pair (y_c, y_r)
Retrieved Context $r(q)$
LLM Backbone
Hidden states
RAG Path
f_{rag} Evidential Head
Dirichlet Params α^{rag}
$\mathcal{L}_{EDL-Total}$
f_{param} Evidential Head
Dirichlet Params α^{param}
Parametric Path
Evidence
Belief Mass
Uncertainty $m^{rag}(\Omega)$
Conflict Score κ
S = {KG, UG}, U = {KN, UN}
$\kappa = m^{param}(S) \cdot m^{rag}(U) + m^{param}(U) \cdot m^{rag}(S)$
Conflict-Aware Weight w_{ds}
$w_{ds} = 1 + \gamma \cdot \kappa \cdot (1 - m^{rag}(\Omega))$
Directed Preference Optimization
$w_{ds} \cdot \mathcal{L}_{DPO} + \mathcal{L}_{SFT}$
Contextual Evidence Quantification (Section 4.1)
Quantifying Knowledge Conflict (Section 4.2)
Uncertainty-Aware Abstention Mechanism (Section 4.3)

Figure 2. Overview of the ERA. The model consists of three components: (1) Contextual Evidence Quantification, where evidential heads project representations from both RAG and Parametric paths into Dirichlet distributions ($\boldsymbol{\alpha}$); (2) Knowledge Conflict Quantification, which utilizes Dempster-Shafer Theory to fuse beliefs and compute a conflict score κ; and (3) Uncertainty-Aware Abstention Mechanism, which modulates the optimization objective based on the detected conflict to mitigate hallucinations.

Knowledge Quadrants and Ideal Alignment. We categorize samples into four distinct knowledge quadrants based on the intersection of internal parametric knowledge ($\mathcal{K}_{\text{param}}$) and external retrieved evidence ($\mathcal{K}_{\text{retr}}$), as illustrated in Figure 1. To ensure trustworthy system behavior, we define specific preference pairs—chosen (y_c) and rejected (y_r) responses—for each quadrant, strictly conditioning the model's operation on the composite knowledge boundary $\mathcal{K}_{\text{rag}}$.

First, we operationalize the boundaries as follows:

- **Determining** $q \in \mathcal{K}_{\textbf{param}}$**:** We sample N answers $\{a_1, \ldots, a_N\}$ from the model $\mathcal{G}(q, \varnothing)$ using different random seeds. If the proportion of correct answers (determined via lexical matching) exceeds a threshold δ, we consider the knowledge internally available:

  $$\frac{1}{N} \sum_{i=1}^{N} \mathbb{I}[C(a_i) = 1] > \delta \quad \text{(Eq. 3)}$$

- **Determining** $q \in \mathcal{K}_{\textbf{retr}}$**:** We employ GPT-4o as a judge with a specialized prompt to assess whether the retrieved passages explicitly contain or imply the ground truth. If the context sufficiently supports the answer, we designate $q \in \mathcal{K}_{\text{retr}}$.

Based on these definitions, we construct the ideal alignment targets for each quadrant:

\footnotesize\arabicenumⓘ **Known-Gold (KG,** $q \in \mathcal{K}_{\textbf{param}} \cap \mathcal{K}_{\textbf{retr}}$**):** Both internal and external knowledge support the correct answer. Since the answer is well-supported, we designate the ground truth as the chosen response y_c. The rejected response y_r is set to "I don't know" (IDK) to penalize unnecessary abstention.

\footnotesize\arabicenumⓘ **Known-Noisy (KN,** $q \in \mathcal{K}_{\textbf{param}} \setminus \mathcal{K}_{\textbf{retr}}$**):** The model possesses the correct answer, but retrieval fails or provides noisy context. This tests *Denoising* capability. Ideally, the model should prioritize internal knowledge. Thus, y_c is the ground truth, while y_r includes context-induced hallucinations and IDK responses to discourage vulnerability to noise and overly conservative behavior.

\footnotesize\arabicenumⓘ **Unknown-Gold (UG,** $q \in \mathcal{K}_{\textbf{retr}} \setminus \mathcal{K}_{\textbf{param}}$**):** The model relies entirely on retrieved documents. This tests *Context Utilization*. The ground truth serves as y_c. To ensure effective knowledge extraction, y_r targets three failure modes: incorrect answers produced despite gold evidence in the provided context, incorrect answers produced without any context that reveal unsuppressed parametric hallucinations, and unjustified IDK responses.

\footnotesize\arabicenumⓘ **Unknown-Noisy (UN,** $q \notin \mathcal{K}_{\textbf{param}} \cup \mathcal{K}_{\textbf{retr}}$**):** Neither source contains the answer. Safety is paramount; the model must abstain. We designate IDK as y_c. Crucially, y_r comprises parametric hallucinations, noise-derived answers, and even the *ground truth* (to prevent unsupported "lucky guesses").

Evaluation Metrics. Table 1 outlines our evaluation framework.We define $\mathcal{S}_{\text{ans}}$ and $\mathcal{S}_{\text{abs}}$ as the sets of queries where the model decides to answer or abstain, respectively. To isolate specific performance behaviors, these prediction sets are intersected with ground-truth partitions, such as answerable queries ($\mathcal{T}_{\text{ans}} = \mathcal{K}_{\text{rag}}$), gold-only contexts ($\mathcal{T}_{\text{gold}} = \mathcal{K}_{\text{retr}} \setminus \mathcal{K}_{\text{param}}$), and noisy contexts ($\mathcal{T}_{\text{noise}} = \mathcal{K}_{\text{param}} \setminus \mathcal{K}_{\text{retr}}$). We employ lexical matching to assess answer correctness, following existing work (Sun et al. 2025). This allows precise quantification of the trade-off between utility (Answer Quality) and reliability (Abstain Quality) via the Overall F1 score.

4. Method

Let $\mathcal{D} = \{(q, y_c, y_r, y_k)\}$ represent the preference alignment dataset, where each tuple contains a query q, a response preference pair (y_c, y_r), and a one-hot encoded knowledge quadrant label y_k. We aim to optimize LLM $\mathcal{G}$ to satisfy the reliability constraints imposed by the preference pairs, while jointly learning to predict the corresponding label y_k.

Figure 2 illustrates the overall workflow. Our proposed framework, ERA, first systematically aggregates evidence scores derived from retrieved contexts to evaluate the reliability and strength of external information (Section 4.1). Subsequently, it computes a conflict score to quantitatively measure the discrepancy between the model's intrinsic parametric knowledge and the provided external evidence (Section 4.2). Guided by this metric, the model dynamically modulates the preference alignment objective to prioritize either robustness or context utilization, thereby ensuring a well-calibrated mapping to the target knowledge quadrant (Section 4.3). These techniques effectively mitigate contextual noise and calibrate the model's behavioral alignment. In the following section, we describe the proposed framework in detail.

4.1. Contextual Evidence Quantification

Standard models typically optimize the conventional cross-entropy loss, which compels them to produce confident point estimates, often leading to uncalibrated overconfidence and conflated epistemic and aleatoric uncertainties. We adopt Evidential Deep Learning (EDL) to disentangle these uncertainties by capturing the predictive distribution density (Sensoy et al. 2018). Moreover, since a single uncertainty estimate in RAG obscures whether uncertainty arises from intrinsic ignorance or knowledge conflicts, we extend EDL with a dual-head architecture to independently quantify evidential support from parametric memory and retrieved context.

Evidential Formulation via Subjective Logic. To distinctively capture the uncertainty from external and internal sources, we integrate two auxiliary evidential heads into the LLM $\mathcal{G}$: f_{rag} and f_{param}. These heads take the hidden states from the retrieval-augmented input and the parametric-only input, respectively, projecting them into logit vectors $z \in \mathbb{R}^4$ corresponding to the four knowledge quadrants. The parametric-only input is formed using the same query without any retrieval context, allowing f_{param} to estimate an evidential prior grounded in parametric memory even when no documents are provided. This prior is defined in the same quadrant-based evidential space as f_{rag}, which enables consistent internal–external comparison and calibration when retrieved evidence is provided.

To disentangle different types of uncertainty, we parameterize the categorical distribution over the quadrants using a Dirichlet prior. This formulation, rooted in Subjective Logic, allows the framework to decompose collected evidence into belief masses for each quadrant and a comprehensive uncertainty term (Jsang 2018). The logit vector z from each head is transformed into non-negative evidence $e \in \mathbb{R}^4$ via the Softplus activation to ensure smooth optimization:

$$e_k = \text{Softplus}(z_k), \quad k \in \{1, \ldots, K\} \quad \text{(Eq. 4)}$$

According to Subjective Logic, the evidence e_k is related to the belief mass b_k and the uncertainty mass u as follows:

$$b_k = \frac{e_k}{S}, \quad u = \frac{K}{S}, \quad \text{where } S = \sum_{k=1}^{K} (e_k + 1) \quad \text{(Eq. 5)}$$

Here, S represents the Dirichlet strength. The sum of all masses is constant: $\sum_{k=1}^{K} b_k + u = 1$. The corresponding Dirichlet distribution parameters are derived as $\alpha_k = e_k + 1$. Intuitively, if the model collects more evidence for a specific quadrant, the corresponding belief mass b_k increases, and the global uncertainty u decreases.

Optimization Objective. To calibrate these evidential outputs, we employ a hybrid objective function that combines the expected risk minimization with a regularizer. Our framework optimizes the objective by integrating cross-entropy loss over the Dirichlet-defined simplex $\text{Dir}(\boldsymbol{p}|\boldsymbol{\alpha})$, thereby minimizing the overall Bayes risk. This yields the analytical form involving the digamma function $\psi(\cdot)$:

$$\mathcal{L}_{\text{fit}}(\boldsymbol{\alpha}, \boldsymbol{y}) = \sum_{k=1}^{K} y_k \big(\psi(S) - \psi(\alpha_k)\big) \quad \text{(Eq. 6)}$$

where $\boldsymbol{y}$ is one-hot target vector. This term encourages the predictive distribution to concentrate around the ground truth quadrant.

While $\mathcal{L}_{\text{fit}}$ encourages correct classification, it does not guarantee that evidence for incorrect classes is suppressed. To mitigate this, we incorporate a Kullback-Leibler (KL) divergence that penalizes the divergence between the predicted posterior and a uniform Dirichlet prior $\text{Dir}(\mathbf{1})$, which represents total uncertainty ($u = 1$).

$$\mathcal{L}_{\text{KL}}(\boldsymbol{\alpha}, \boldsymbol{y}) = \text{KL}\big[\text{Dir}(\tilde{\boldsymbol{\alpha}}) \| \text{Dir}(\mathbf{1})\big] \quad \text{(Eq. 7)}$$

Here, $\tilde{\boldsymbol{\alpha}} = \boldsymbol{y}(1 - \boldsymbol{y}) \odot \boldsymbol{\alpha}$ represents the adjusted parameters used to penalize misleading evidence assigned to non-target classes.

The individual evidential loss for a given head is computed as a weighted sum, where λ_t is an annealing coefficient that linearly increases over training steps t to prevent premature convergence:

$$\mathcal{L}_{\text{EDL}}(\boldsymbol{\alpha}, \boldsymbol{y}) = \mathcal{L}_{\text{fit}}(\boldsymbol{\alpha}, \boldsymbol{y}) + \lambda_t \mathcal{L}_{\text{KL}}(\boldsymbol{\alpha}, \boldsymbol{y}) \quad \text{(Eq. 8)}$$

Finally, since our framework relies on two distinct evidential sources to detect knowledge conflicts, we apply this objective to the outputs of both heads. Let $\boldsymbol{\alpha}^{\text{rag}}$ and $\boldsymbol{\alpha}^{\text{param}}$ denote the Dirichlet parameters derived from f_{rag} and f_{param}, respectively. The total evidential learning objective is defined as the sum of losses for both contexts:

$$\mathcal{L}_{\text{EDL-Total}} = \mathcal{L}_{\text{EDL}}(\boldsymbol{\alpha}^{\text{rag}}, \boldsymbol{y}) + \mathcal{L}_{\text{EDL}}(\boldsymbol{\alpha}^{\text{param}}, \boldsymbol{y}) \quad \text{(Eq. 9)}$$

4.2. Quantifying Knowledge Conflict

Having independently quantified the evidential distributions from parametric and retrieval sources, we next assess their alignment. Detecting discrepancies where the model's internal priors contradict retrieved evidence is critical, as such conflicts are often precursors to hallucinations. To rigorously measure this discord, we employ Dempster-Shafer Theory (DST) to fuse the distinct evidential sources. This fusion process yields a conflict score κ, which serves as a robust proxy for identifying hallucinations and delineating the model's knowledge boundaries.

Belief Mass Reduction. While our evidential heads predict fine-grained knowledge quadrants, the fundamental conflict in RAG scenarios stems from the validity of the evidence: whether the source provides sufficient information to support an answer or fails to do so (due to noise or ignorance). To capture this from a RAG perspective, we aggregate the quadrant evidence into two high-level hypotheses: Supported ($\mathcal{S}$) and Unsupported ($\mathcal{U}$).

$$\mathcal{S} = \{\text{KG}, \text{UG}\}, \quad \mathcal{U} = \{\text{KN}, \text{UN}\} \quad \text{(Eq. 10)}$$

For each source $s \in \{\text{rag}, \text{param}\}$, corresponding to the evidential outputs of heads f_{rag} and f_{param} respectively, we reduce the 4-dimensional Dirichlet parameters into a 2-dimensional belief mass assignment. The mass functions for these hypotheses and the global uncertainty (Ω) are derived by summing the belief masses of the constituent quadrants:

$$m^s(\mathcal{S}) = \sum_{k \in \mathcal{S}} b^s_k, \quad m^s(\mathcal{U}) = \sum_{k \in \mathcal{U}} b^s_k, \quad m^s(\Omega) = u^s \quad \text{(Eq. 11)}$$

This reduction allows us to focus on the semantic validity of the evidence—specifically, whether the parametric memory and the retrieved context support the generation of an answer or remain unsupported—while strictly preserving the global uncertainty $m^s(\Omega)$.

Conflict Score (κ**) Calculation.** We utilize Dempster's Rule of Combination to rigorously quantify the tension between the model's internal beliefs (m^{param}) and the external evidence (m^{rag}). The conflict score κ represents the probability mass assigned to the empty set, arising when sources support mutually exclusive hypotheses:

$$\kappa = m^{\text{param}}(\mathcal{S}) \cdot m^{\text{rag}}(\mathcal{U}) + m^{\text{param}}(\mathcal{U}) \cdot m^{\text{rag}}(\mathcal{S}) \quad \text{(Eq. 12)}$$

Crucially, this formulation inherently accounts for uncertainty. Since the sum of masses is fixed to 1, a high uncertainty $m(\Omega)$ in either source naturally reduces the masses assigned to $\mathcal{S}$ and $\mathcal{U}$, thereby suppressing κ. Thus, a high κ value indicates a critical dissonance where one source strongly supports the answer while the other strongly refutes it.

4.3. Uncertainty-Aware Abstention Mechanism

The ultimate goal of our framework is to calibrate the model's behavior so that it selectively abstains when intrinsic knowledge is insufficient or contradicts reliable external evidence. Standard preference optimization (e.g., DPO) treats all training samples uniformly, which is suboptimal for learning these delicate decision boundaries (Rafailov et al. 2023). To address this, we propose a comprehensive training strategy that prioritizes learning from high-conflict, high-evidence scenarios.

Conflict-Aware Loss Modulation. We hypothesize that the most critical signal for learning robustness lies in samples where the model's internal belief conflicts with the retrieved context—but only if that context is trustworthy. We formulate a dynamic weighting factor w_{ds} applied to the alignment loss:

$$w_{\text{ds}} = 1 + \gamma \cdot \kappa \cdot (1 - m^{\text{rag}}(\Omega)) \quad \text{(Eq. 13)}$$

Here, γ is a hyperparameter controlling the sensitivity of the modulation, and the term $(1 - m^{\text{rag}}(\Omega))$ acts as a reliability gate. Specifically, when the conflict κ is high and the retrieval uncertainty is low, the weight w_{ds} increases, forcing the model to focus on resolving the dissonance—typically by aligning with strong external evidence or correcting hallucinations. Conversely, when the retrieval itself is uncertain ($m^{\text{rag}}(\Omega) \approx 1$), this gate effectively closes, keeping w_{ds} near 1.0. This mechanism prevents the model from aggressively updating its policy based on ambiguous or noisy retrieval results.

Preference Alignment. To align the model with human preferences, we employ Direct Preference Optimization (DPO), which optimizes the policy directly using pairs of chosen (y_c) and rejected (y_r) responses without requiring a separate reward model. Given a query q and retrieved context $r(q)$, the DPO loss is formulated as:

$$\mathcal{L}_{\text{DPO}}(\pi_\theta, \pi_{\text{ref}}) = -\log \sigma\left(\tau \left[\log \frac{\pi_\theta(y_c|q, r(q))}{\pi_{\text{ref}}(y_c|q, r(q))} - \log \frac{\pi_\theta(y_r|q, r(q))}{\pi_{\text{ref}}(y_r|q, r(q))}\right]\right), \quad \text{(Eq. 14)}$$

where π_θ is the policy being trained, π_{ref} is the frozen reference model, and τ is the temperature parameter.

However, DPO primarily focuses on degrading the likelihood of rejected responses rather than actively improving the chosen ones. To mitigate this and maintain generation coherence, we add a supervised fine-tuning (SFT) auxiliary loss on the chosen responses:

$$\mathcal{L}_{\text{SFT}}(\pi_\theta) = -\frac{1}{T} \sum_{t=1}^{T} \log \pi_\theta(y_{c,t} \,|\, q, r(q), y_{c,<t}), \quad \text{(Eq. 15)}$$

where $y_{c,t}$ denotes the t-th token of the chosen response. This auxiliary term explicitly reinforces the model's ability to generate high-quality preferred outputs.

Unified Training Objective. We integrate the evidential learning and the preference alignment into a unified objective. The final total loss $\mathcal{L}_{\text{total}}$ is a weighted combination of the conflict-modulated DPO loss, the total evidential loss, and the SFT auxiliary loss:

$$\mathcal{L}_{\text{total}} = w_{\text{ds}} \cdot \mathcal{L}_{\text{DPO}}(\pi\theta, \pi_{\text{ref}}) + \mathcal{L}_{\text{EDL-Total}} + \mathcal{L}_{\text{SFT}} \quad \text{(Eq. 16)}$$

where w_{ds} is the conflict-aware modulation factor derived in Eq. 13.

Table 2. Performance comparison across three QA datasets using Qwen3-8B and Llama3-8B backbones. Metrics are categorized into: **Answer Quality**, **Retrieval Handling**, and **Abstention Quality**. **Overall F1** represents the harmonic mean balancing answer generation and abstention.

Method	Answer Quality			Retrieval Handling		Abstention Quality			Overall F1
	Recall	Prec.	F1	Denoise	Context	Recall	Prec.	F1	
Qwen3-8B-Base									
Logprob (t =−2)	0.308	0.933	0.463	0.303	0.221	0.050	0.564	0.091	0.153
Logprob (t =−1)	0.197	0.933	0.325	0.187	0.144	0.689	0.360	0.473	0.385
ICL	0.178	0.130	0.150	0.184	0.142	0.048	0.404	0.085	0.109
Self-Consistency	0.203	0.415	0.273	0.219	0.142	0.777	0.350	0.483	0.349
p_{true} ($t = 0.5$)	0.209	0.931	0.342	0.245	0.128	0.521	0.399	0.452	0.389
p_{true} ($t = 0.7$)	0.133	0.973	0.234	0.161	0.036	0.883	0.353	0.504	0.320
p_{true} ($t = 0.9$)	0.014	**1.000**	0.029	0.014	0.001	**0.990**	0.294	0.453	0.054
DTA	0.436	0.410	0.422	0.380	0.319	0.476	0.566	0.517	0.465
ERA	**0.622**	0.532	**0.573**	**0.606**	**0.454**	0.408	**0.715**	**0.520**	**0.545**
Llama3.1-8B									
Logprob (t =−2)	0.509	0.910	**0.653**	**0.568**	**0.392**	0.314	0.424	0.361	0.465
Logprob (t =−1)	0.282	0.914	0.431	0.346	0.199	0.775	0.340	0.472	0.451
ICL	0.474	0.336	0.394	0.548	0.326	0.014	**0.800**	0.027	0.050
Self-Consistency	0.386	0.492	0.432	0.392	0.300	0.559	0.368	0.444	0.438
p_{true} ($t = 0.5$)	0.364	0.931	0.523	0.375	0.235	0.311	0.556	0.268	0.312
p_{true} ($t = 0.7$)	0.337	0.934	0.495	0.352	0.207	0.497	0.484	0.293	0.368
p_{true} ($t = 0.9$)	0.253	**0.952**	0.400	0.300	0.130	**0.779**	0.402	0.222	0.255
DTA	0.490	0.544	0.516	0.510	0.322	0.596	0.481	0.533	0.524
ERA	**0.537**	0.680	0.600	0.504	0.367	0.759	0.506	**0.607**	**0.604**

Optimizing this objective, ERA jointly learns to adhere to human preferences while simultaneously calibrating its predictive distributions to accurately reflect knowledge boundaries and uncertainty.

5. Experiment

We evaluate ERA on multiple RAG datasets and perform various analyses to assess how our framework improves calibration quality. Due to space constraints, further details, including implementation details and results from varying LLMs, are provided in the Appendix.

5.1. Performance Evaluation

Datasets. Consistent with prior RAG studies (Fang et al. 2024; Sun et al. 2025), we validate the effectiveness of our proposed framework across diverse question-answering scenarios by conducting comprehensive evaluations on three widely established open-domain QA benchmarks: Natural Questions (NQ) (Kwiatkowski et al. 2019), TriviaQA (Joshi et al. 2017), and WebQuestions (WebQ) (Berant et al. 2013). These datasets were selected to cover a broad spectrum of query types, ranging from real-world user search queries (NQ) to complex trivia-based reasoning (TriviaQA) and entity-centric questions (WebQ). For retrieval, we utilize Dense Passage Retrieval (DPR) (Karpukhin et al. 2020) to obtain the top-3 passages from Wikipedia, assessing knowledge availability by verifying the presence of the ground truth answer within the retrieved contexts.

To evaluate generalization capabilities on information unseen during pre-training, we construct the Wiki Event benchmark by parsing the 'Events' sections of Wikipedia annual pages ranging from 2010 to 2025. This temporal coverage specifically targets events post-dating standard knowledge cutoffs. For specific event descriptions, we scrape the referenced external news articles to establish them as the gold contexts. We then leverage GPT-4o to generate factual question-answer pairs derived directly from these Wikipedia summaries. Crucially, we also synthesize counterfactual contexts by systematically perturbing key entities in the gold contexts to simulate realistic scenarios of noisy or conflicting retrieval.

Baselines. We implement five calibration strategies as baselines: (1) Logprob calculates the average log-probability of all tokens in the generated answer, serving as a basic confidence metric (Kadavath et al. 2022). (2)

ICL prompts the model to verbally self-evaluate the correctness of its answer (Wei et al. 2022). (3) Self-Consistency computes confidence based on the agreement rate among multiple sampled outputs for the same query (Wang et al. 2023). (4) P_{true} extracts the specific softmax probability assigned to the "True" token given a verification prompt, providing a continuous score (Kadavath et al. 2022). (5) DTA employs DPO to train models to abstain from answering when retrieved evidence is insufficient, using curated abstention datasets (Sun et al. 2025).

Results. Table 2 presents a performance comparison of calibration strategies regarding Answer Quality, Retrieval Handling, and Abstention Quality. ERA achieves the highest Overall F1 scores across both Llama3-8B (Dubey et al. 2024) and Qwen3-8B-Base (Yang et al. 2025) backbones. While confidence based methods (e.g., Logprob, P_{true}) falter due to threshold sensitivity and inability to distinguish uncertainty types, and supervised baselines like DTA are prone to memorizing specific refusal patterns rather than learning robust decision boundaries, our evidential framework overcomes these limitations. By explicitly quantifying belief masses to identify "true unknowns" (epistemic uncertainty), ERA delivers superior Abstention Precision, effectively balancing answer generation and abstention without succumbing to the overfitting or calibration pitfalls observed in prior baselines.

Figure 3 demonstrates that ERA significantly outperforms the strongest baseline (DTA) on the out-of-distribution Wiki Event dataset, achieving a superior Overall F1. While DTA achieves high Abstention Recall, its critically low Answer F1 exposes a vulnerability to overfitting: the model appears to memorize specific refusal patterns from the training data, leading to indiscriminate over-abstention on unseen events. In contrast, ERA effectively mitigates this overfitting by relying on disentangled uncertainty estimates, doubling the Answer F1 while maintaining high Abstention F1, thereby validating its robust generalization capability.

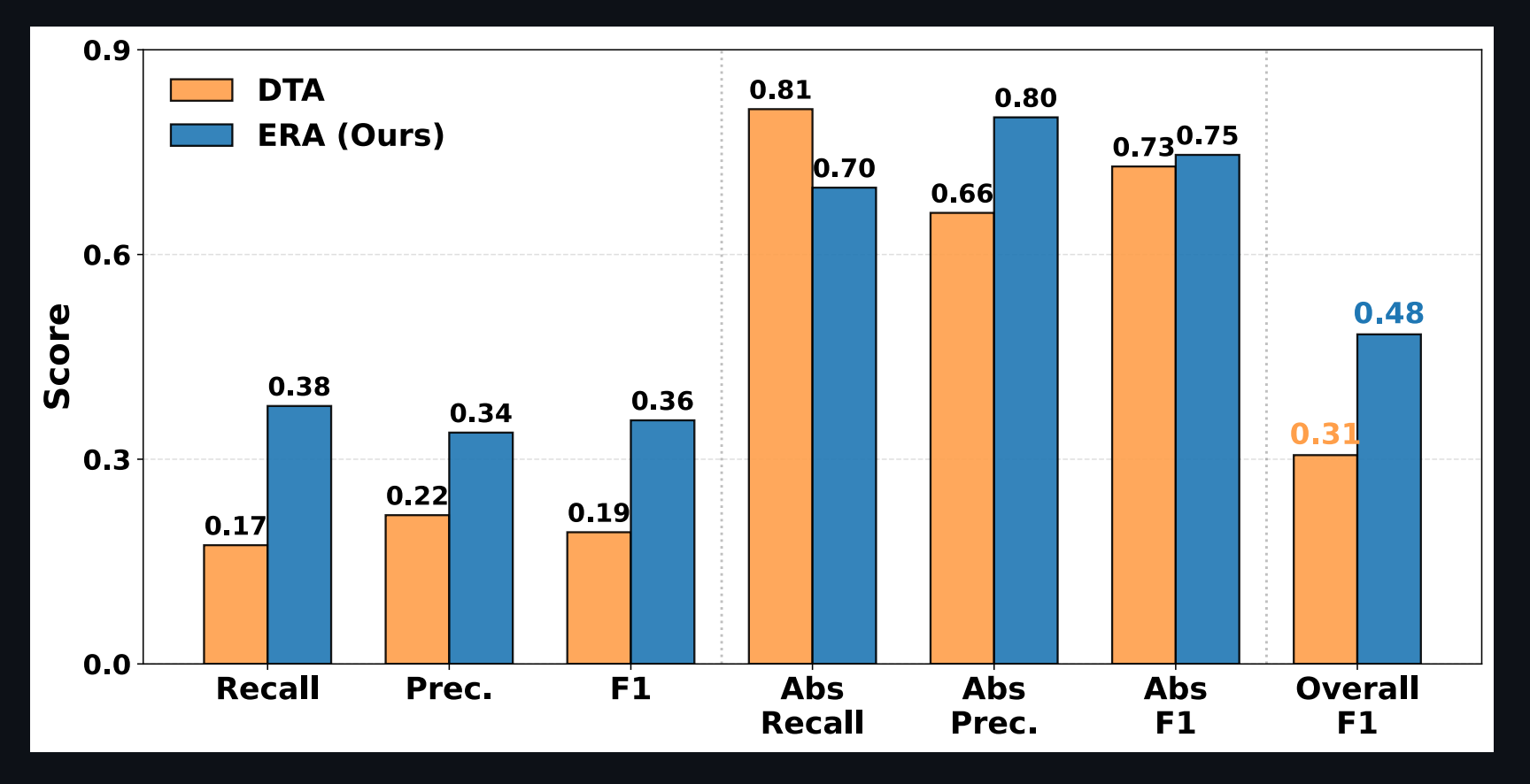
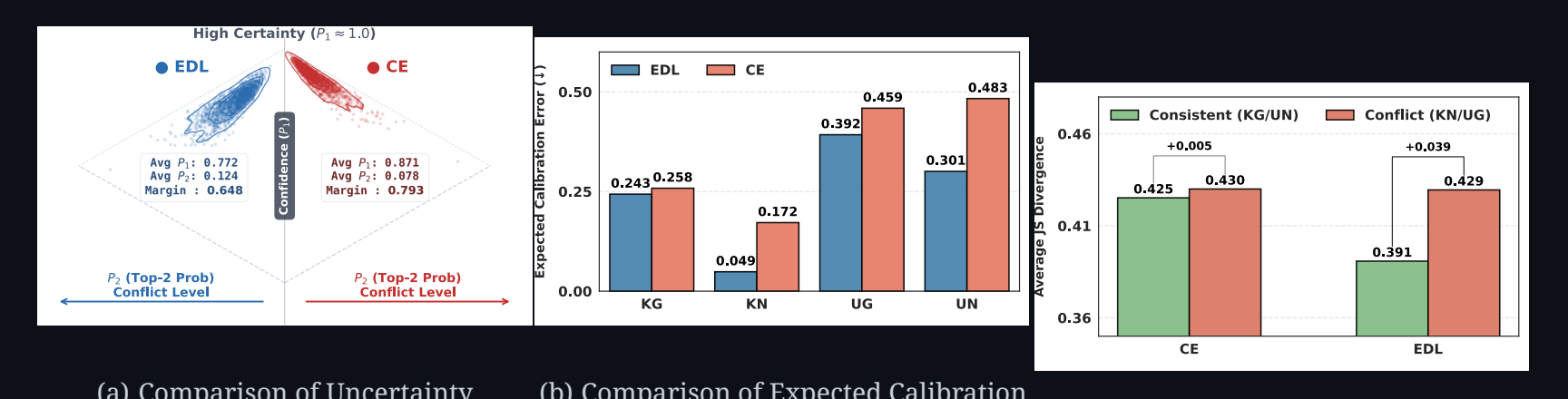


Figure 3. Generalizability evaluation on the Wiki Event dataset using Qwen3-8B-Base.

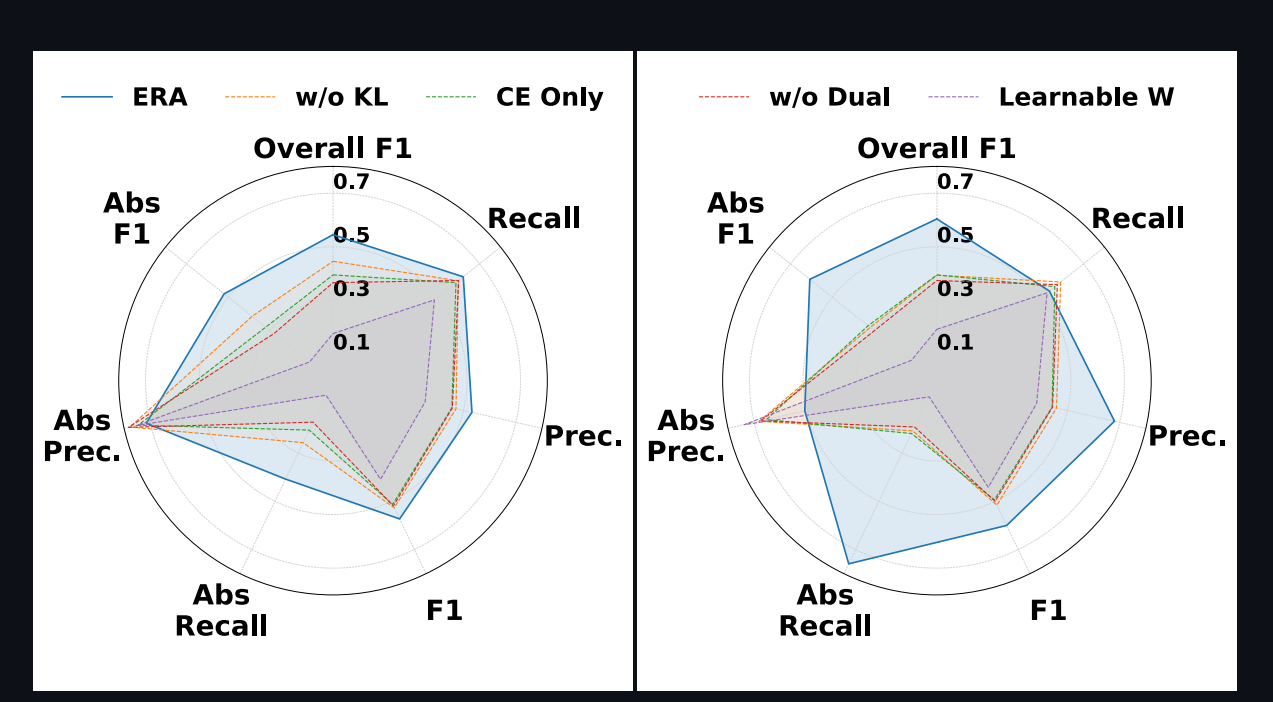


(a) Comparison of Uncertainty Distributions

(b) Comparison of Expected Calibration Error

(c) Conflict Sensitivity Analysis

Figure 4. Quantitative analysis of uncertainty. This figure compares the proposed Evidential Deep Learning (EDL) model against the Cross-Entropy (CE) baseline across three dimensions: (a) A visualization of prediction probability distributions in conflict situations (KN, UG); (b) A comparison of Expected Calibration Error (ECE) across data types; and (c) An evaluation of model sensitivity via the Jensen-Shannon Divergence (JSD) between consistent and conflicting information.



(a) Qwen3-8B-Base

(b) Llama3.1-8B

Figure 5. Performance comparison of ablation studies. The consistent degradation in Overall F1 scores observed when removing or altering individual components validates the distinct contribution of each module in the ERA framework.

5.2. Component Analysis

Ablation Study. We conduct an ablation study to evaluate the contribution of each component in our full model (ERA). The following variations are compared: (1) w/o Dual removes the parametric memory head, relying solely on RAG context and thereby bypassing the internal-external conflict detection mechanism; (2) Learnable W replaces the evidence-based Dempster-Shafer fusion with a simple learnable scalar gate to weight the loss; (3) CE Only substitutes the evidential learning objective with standard cross-entropy loss, ignoring explicit uncertainty quantification; and (4) w/o KL omits the KL divergence regularization term from the evidential loss, utilizing only the data fit term.

Figure 5 shows that the full model with all components performs the best across all metrics. Specifically, the removal of the dual-head architecture (w/o Dual) results in a critical drop in Abstain F1, confirming that explicitly modeling the conflict between parametric and retrieved knowledge is essential for mitigating hallucinations. Furthermore, the Learnable W variant exhibits the lowest Trade-Off scores, indicating that simple scalar gating is insufficient for integrating conflicting evidence compared to our evidential Dempster-Shafer fusion. Finally, the sub-optimal performance of the CE baseline and the ablation without KL regularization highlights the necessity of evidential uncertainty quantification. Specifically, KL regularization is essential for effectively distinguishing true unknowns from data ambiguity, thereby ensuring robust calibration.

Qualitative Analysis. We first investigate how well our evidential framework calibrates uncertainty compared to standard optimization methods. To visualize this, we compare the uncertainty distributions and Expected Calibration Error (ECE) between the Cross-Entropy (CE) baseline and our Evidential Deep Learning (EDL) approach. Figure 4(a) illustrates the simplex of prediction probabilities, where CE (red) tends to cluster densely around high-confidence regions (Avg $P_1 \approx 0.87$), indicating a strong tendency toward overconfidence even when the model is likely wrong. In contrast, EDL (blue) exhibits a more dispersed distribution with a lower average confidence (Avg $P_1 \approx 0.77$), suggesting a more conservative and reliable estimation of belief. This improved calibration is quantitatively supported by Figure 4(b), where EDL consistently achieves substantially lower ECE scores across all four knowledge quadrants (KG, KN, UG, UN). Notably, in the high-risk Unknown-Noisy (UN) and Known-Noisy (KN) scenarios, EDL significantly reduces calibration error, demonstrating its effectiveness in mitigating the overconfidence typically observed in hallucinating models.

Next, we examine the model's sensitivity to knowledge conflicts—specifically, determining whether the system can effectively distinguish between scenarios where its internal parametric memory aligns with external retrieved evidence versus when they contradict. To quantify this, we utilize the Jensen-Shannon (JS) divergence to measure the distributional discrepancy between the predictive distributions derived from the parametric and retrieval-augmented paths. Figure 4(c) reveals that the CE baseline fails to capture this distinction, showing a negligible increase in divergence (+0.005) when a conflict occurs. Conversely, our method demonstrates a marked sensitivity to discordance, with a significant increase in JS divergence (+0.039) for conflicting samples compared to consistent ones. This suggests that our evidential framework effectively disentangles aleatoric uncertainty from epistemic belief, allowing the model to explicitly detect conflicts and thereby filter out contaminated knowledge more effectively than standard probabilistic approaches.

6. Conclusion

This paper proposes ERA, a principled framework designed to address the critical challenge of knowledge conflicts in RAG systems. We propose shifting the paradigm of confidence estimation from scalar probabilities to evidence distributions via Evidential Deep Learning and introduced a rigorous geometric quantification of "belief conflict" using Dempster-Shafer Theory. With these components, our method successfully disentangles epistemic uncertainty from aleatoric ambiguity and explicitly measures the discordance between its internal parametric knowledge and external retrieved evidence. As a result, our method produces well-calibrated confidence estimates and significantly outperforms existing baselines in reducing hallucination risks while maintaining high accuracy on answerable queries. We hope that these findings underscore the necessity of modeling evidence distributions, laying the groundwork for future research into trustworthy AI systems.

References

(1)

Asai et al. (2024) Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2024. Self-RAG: Learning to Retrieve, Generate, and Critique Through Self-Reflection. In *Proceedings of the International Conference on Learning Representations*.

Berant et al. (2013) Jonathan Berant, Andrew Chou, Roy Frostig, and Percy Liang. 2013. Semantic Parsing on Freebase from Question-Answer Pairs. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*. 1533–1544.

Charpentier et al. (2020) Bertrand Charpentier, Daniel Zügner, and Stephan Günnemann. 2020. Posterior Network: Uncertainty Estimation Without OOD Samples via Density-Based Pseudo-Counts. In *Advances in Neural Information Processing Systems*, Vol. 33. 1356–1367.

Chen et al. (2024) Lu Chen, Ruqing Zhang, Jiafeng Guo, Yixing Fan, and Xueqi Cheng. 2024. Controlling Risk of Retrieval-Augmented Generation: A Counterfactual Prompting Framework. In *Findings of the Association for Computational Linguistics: EMNLP*. 2380–2393.

Chuang et al. (2024) Yung-Sung Chuang, Yujia Xie, Hongyin Luo, Yoon Kim, James R Glass, and Pengcheng He. 2024. DoLa: Decoding by Contrasting Layers Improves Factuality in Large Language Models. In *Proceedings of the International Conference on Learning Representations*.

Dubey et al. (2024) Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. 2024. The Llama 3 Herd of Models. *arXiv preprint arXiv:2407.21783* (2024).

Fang et al. (2024) Feiteng Fang, Yuelin Bai, Shiwen Ni, Min Yang, Xiaojun Chen, and Ruifeng Xu. 2024. Enhancing Noise Robustness of Retrieval-Augmented Language Models with Adaptive Adversarial Training. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*. 10028–10039.

Gal and Ghahramani (2016) Yarin Gal and Zoubin Ghahramani. 2016. Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning. In *Proceedings of the International Conference on Machine Learning*. PMLR, 1050–1059.

Guerreiro et al. (2023) Nuno M Guerreiro, Elena Voita, and André FT Martins. 2023. Looking for a Needle in a Haystack: A Comprehensive Study of Hallucinations in Neural Machine Translation. In *Proceedings of the Conference of the European Chapter of the Association for Computational Linguistics*. 1059–1075.

Guo et al. (2017) Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. 2017. On Calibration of Modern Neural Networks. In *Proceedings of the International Conference on Machine Learning*. PMLR, 1321–1330.

Hadji-Kyriacou and Arandjelović (2024) Avelina A Hadji-Kyriacou and Ognjen Arandjelović. 2024. Would I Lie to You? Inference Time Alignment of Language Models Using Direct Preference Heads. In *Advances in Neural Information Processing Systems*, Vol. 37. 95380–95405.

Han et al. (2022) Zongbo Han, Changqing Zhang, Huazhu Fu, and Joey Tianyi Zhou. 2022. Trusted Multi-View Classification with Dynamic Evidential Fusion. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45, 2 (2022), 2551–2566.

Jiang et al. (2023) Zhengbao Jiang, Frank F Xu, Luyu Gao, Zhiqing Sun, Qian Liu, Jane Dwivedi-Yu, Yiming Yang, Jamie Callan, and Graham Neubig. 2023. Active Retrieval Augmented Generation. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*. 7969–7992.

Jin et al. (2025) Mingyu Jin, Weidi Luo, Sitao Cheng, Xinyi Wang, Wenyue Hua, Ruixiang Tang, William Yang Wang, and Yongfeng Zhang. 2025. Disentangling Memory and Reasoning Ability in Large Language Models. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*. 1681–1701.

Joshi et al. (2017) Mandar Joshi, Eunsol Choi, Daniel S Weld, and Luke Zettlemoyer. 2017. TriviaQA: A Large Scale Distantly Supervised Challenge Dataset for Reading Comprehension. *arXiv preprint arXiv:1705.03551* (2017).

Jsang (2018) Audun Jsang. 2018. *Subjective Logic: A Formalism for Reasoning Under Uncertainty*. Springer Publishing Company, Incorporated.

Kadavath et al. (2022) Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. 2022. Language Models (Mostly) Know What They Know. *arXiv preprint arXiv:2207.05221* (2022).

Kalai et al. (2025) Adam Tauman Kalai, Ofir Nachum, Santosh S Vempala, and Edwin Zhang. 2025. Why Language Models Hallucinate. *arXiv preprint arXiv:2509.04664* (2025).

Karpukhin et al. (2020) Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. Dense Passage Retrieval for Open-Domain Question Answering. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*. 6769–6781.

Kuhn et al. (2023) Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. 2023. Semantic Uncertainty: Linguistic Invariances for Uncertainty Estimation in Natural Language Generation. In *Proceedings of the International Conference on Learning Representations*.

Kwiatkowski et al. (2019) Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, et al. 2019. Natural Questions: A Benchmark for Question Answering Research. *Transactions of the Association for Computational Linguistics* 7 (2019), 453–466.

Lakshminarayanan et al. (2017) Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. 2017. Simple and Scalable Predictive Uncertainty Estimation Using Deep Ensembles. In *Advances in Neural Information Processing Systems*, Vol. 30.

Lewis et al. (2020) Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In *Advances in Neural Information Processing Systems*, Vol. 33. 9459–9474.

Lin et al. (2022) Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. TruthfulQA: Measuring How Models Mimic Human Falsehoods. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*. 3214–3252.

Longpre et al. (2021) Shayne Longpre, Kartik Perisetla, Anthony Chen, Nikhil Ramesh, Chris DuBois, and Sameer Singh. 2021. Entity-Based Knowledge Conflicts in Question Answering. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*.

Mallen et al. (2023) Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. 2023. When Not to Trust Language Models: Investigating Effectiveness of Parametric and Non-Parametric Memories. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*. 9802–9822.

Maynez et al. (2020) Joshua Maynez, Shashi Narayan, Bernd Bohnet, and Ryan McDonald. 2020. On Faithfulness and Factuality in Abstractive Summarization. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*. 1906–1919.

Niu et al. (2024) Cheng Niu, Yuanhao Wu, Juno Zhu, Siliang Xu, Kashun Shum, Randy Zhong, Juntong Song, and Tong Zhang. 2024. RagTruth: A Hallucination Corpus for Developing Trustworthy Retrieval-Augmented Language Models. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*. 10862–10878.

Park et al. (2025) Sungwon Park, Sungwon Han, and Meeyoung Cha. 2025. Enhancing Domain Generalization for Robust Machine-Generated Text Detection. *IEEE Transactions on Knowledge and Data Engineering* (2025).

Park et al. (2021) Sungwon Park, Sungwon Han, Sundong Kim, Danu Kim, Sungkyu Park, Seunghoon Hong, and Meeyoung Cha. 2021. Improving unsupervised image clustering with robust learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 12278–12287.

Rafailov et al. (2023) Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. 2023. Direct Preference Optimization: Your Language Model is Secretly a Reward Model. In *Advances in Neural Information Processing Systems*, Vol. 36. 53728–53741.

Sensoy et al. (2018) Murat Sensoy, Lance Kaplan, and Melih Kandemir. 2018. Evidential Deep Learning to Quantify Classification Uncertainty. In *Advances in Neural Information Processing Systems*, Vol. 31.

Sensoy et al. (2021) Murat Sensoy, Maryam Saleki, Simon Julier, Reyhan Aydogan, and John Reid. 2021. Misclassification Risk and Uncertainty Quantification in Deep Classifiers. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. 2484–2492.

Shin and Kim (2025) SungUk Shin and Youngjoon Kim. 2025. Enhancing graph of thought: Enhancing prompts with LLM rationales and dynamic temperature control. In *The Thirteenth International Conference on Learning Representations*.

Sun et al. (2025) Xin Sun, Jianan Xie, Zhongqi Chen, Qiang Liu, Shu Wu, Yuehe Chen, Bowen Song, Zilei Wang, Weiqiang Wang, and Liang Wang. 2025. Divide-Then-Align: Honest Alignment Based on the Knowledge Boundary of RAG. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*. 11461–11480.

Wang et al. (2023) Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V Le, Ed H Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-Consistency Improves Chain of Thought Reasoning in Language Models. In *Proceedings of the International Conference on Learning Representations*.

Wei et al. (2022) Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. 2022. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. In *Advances in Neural Information Processing Systems*, Vol. 35. 24824–24837.

Wen et al. (2024) Bingbing Wen, Bill Howe, and Lucy Lu Wang. 2024. Characterizing LLM Abstention Behavior in Science QA with Context Perturbations. *arXiv preprint arXiv:2404.12452* (2024).

Wu et al. (2024) Di Wu, Jia-Chen Gu, Fan Yin, Nanyun Peng, and Kai-Wei Chang. 2024. Synchronous Faithfulness Monitoring for Trustworthy Retrieval-Augmented Generation. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*. 9390–9406.

Yang et al. (2025) An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. 2025. Qwen3 Technical Report. *arXiv preprint arXiv:2505.09388* (2025).

Zhang et al. (2024) Muru Zhang, Ofir Press, William Merrill, Alisa Liu, and Noah A Smith. 2024. How Language Model Hallucinations Can Snowball. In *Proceedings of the International Conference on Machine Learning*. 59670–59684.

Appendix A Appendix

A.1. Training Details

We release the source code and dataset for ERA at `https://anonymous.4open.science/r/ERA-318B`.

Dataset Details. To rigorously evaluate our framework, we first construct a dataset categorized by the interplay between the model's internal knowledge state and the quality of external retrieval. We process standard QA datasets using the target LLM to establish its parametric knowledge boundary, classifying each query as Known or Unknown based on the model's ability to answer correctly without retrieval. To determine membership in $\mathcal{K}_{\text{param}}$, we set the consistency threshold δ to 1.0 using $N = 10$ sampled responses. Subsequently, for each query, we retrieve a pool of candidate contexts containing a mix of relevant ("golden") and irrelevant ("noisy") passages. Based on the presence of golden contexts in the top-3 retrieved results, we stratify the data into four distinct quadrants. Following prior work, we randomly sample 5,000 examples for training and 3,000 for evaluation from these stratified quadrants across three datasets, setting the IDK-ratio hyperparameter to 0.7 for the training distribution.

To further assess the generalizability of RAG models against noisy retrieval contexts, we constructed a synthetic dataset derived from the Wiki Event corpus. We employed GPT-4o to generate context-dependent questions and short answers directly from the original reference documents. To simulate realistic and challenging retrieval scenarios, each query is paired with a stratified pool of 10 contexts designed to represent varying levels of relevance. This pool comprises a single Golden Context containing the ground truth, four Hard Negatives generated by the LLM to exhibit high topical similarity and keyword overlap while explicitly excluding the correct answer, and five Easy Negatives randomly sampled from the global corpus to represent irrelevant noise. This hierarchical structure allows for a granular assessment of the model's ability to discern correct evidence amidst varying degrees of semantic interference and distraction.

Implementation Details. We employ Qwen3-8B-Base and Llama-3.1-8B as the primary backbone models, optimized via 4-bit Quantized LoRA (QLoRA) with nf4 quantization and bfloat16 precision to ensure computational efficiency. LoRA adapters are applied to all linear layers with a rank $r = 8$, $\alpha = 16$, and a dropout rate of 0.05. We train the model for 1 epoch using the AdamW optimizer with a cosine learning rate scheduler, initialized at $5e-5$ with a warmup ratio of 0.1. For the ERA framework, we introduce two lightweight evidential heads that map hidden states to 4-dimensional Dirichlet parameters using a Softplus activation. We utilize the Digamma loss with KL divergence regularization, applying an annealing schedule over the first $2{,}000$ steps for stability. The balancing coefficients λ_{rag} and λ_{param} are set to 0.1, while the conflict-aware weighting factor γ for Dempster-Shafer combination is set to 1.0. Additionally, a standard SFT auxiliary loss with a coefficient of 1.0 is incorporated to maintain generation quality. All experiments are performed on 4 NVIDIA A100 GPUs using PyTorch.

A.2. Further Results on Honest Performance

Due to space limitations in the main manuscript, we present additional results here.

Generalization Analysis. As illustrated in Figure 6, our evaluation on Llama-3.1-8B highlights the superior generalization capability of ERA. While baselines like DTA achieve high abstention recall, they demonstrate a tendency toward over-conservatism, indiscriminately rejecting answerable In-Distribution (ID) queries alongside OOD noise. In contrast, ERA establishes a precise decision boundary that effectively distinguishes between ID and OOD inputs. By strictly refusing OOD inputs without compromising the ability to answer valid queries, ERA achieves a marked improvement in overall F1 scores, demonstrating robust generalization across mixed distributions.

Add-on to RAFT. The performance advantage extends to models explicitly optimized for Retrieval Augmented Fine-Tuning (RAFT), such as ChatQA. As shown in Figure 7, applying ERA to these architectures yields substantial improvements in standard performance metrics compared to DTA. Building upon their pre-training for retrieval contexts, integrating ERA further refines their ability to discriminate between relevant and irrelevant information.

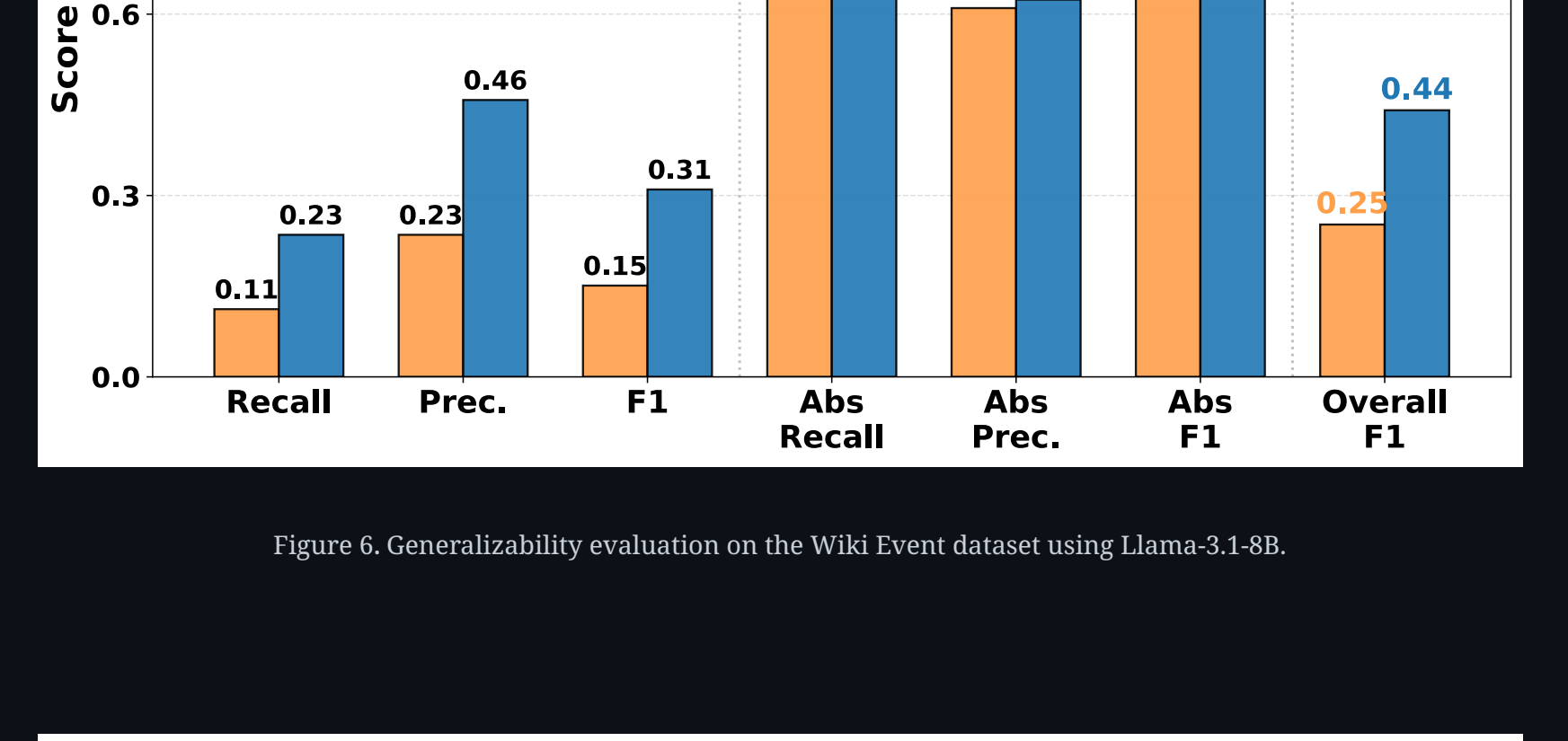


Figure 6. Generalizability evaluation on the Wiki Event dataset using Llama-3.1-8B.

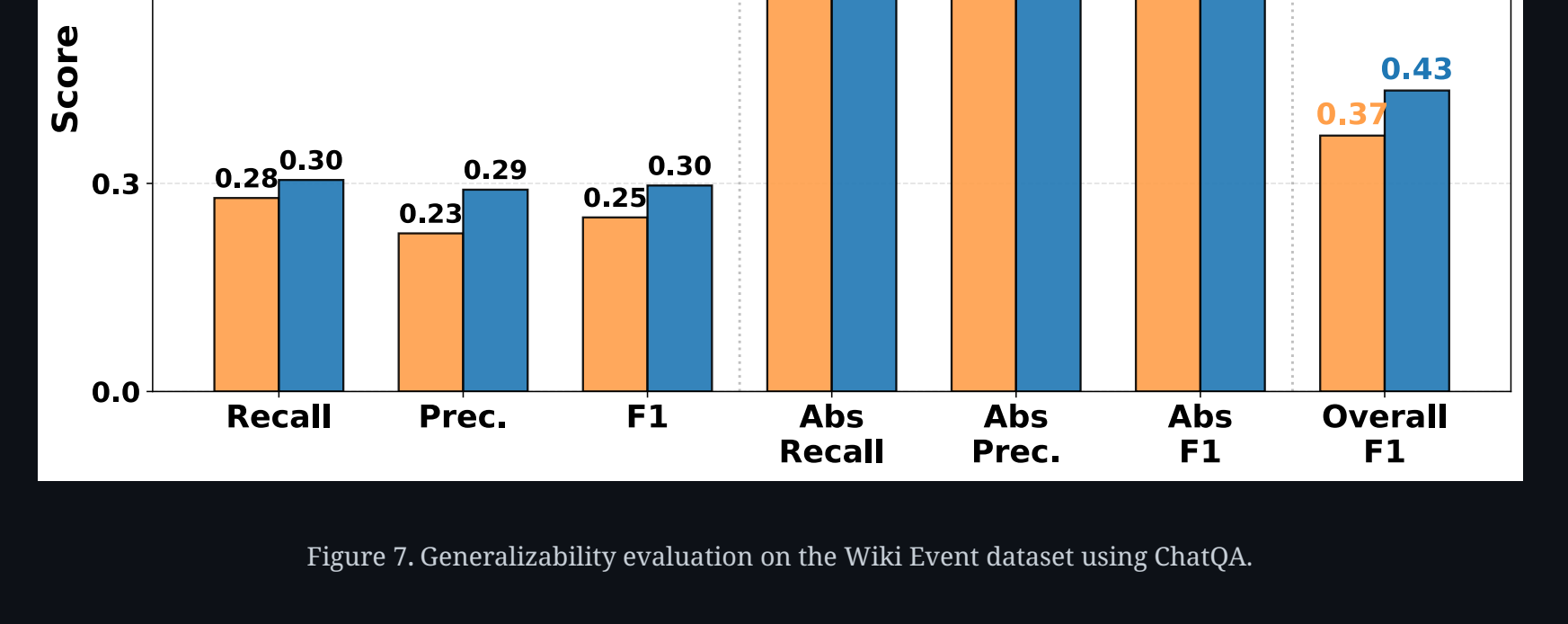


Figure 7. Generalizability evaluation on the Wiki Event dataset using ChatQA.

Feeling lucky? Conversion report (W) Report an issue View original on arXiv

Copyright Privacy Policy Generated by LaTeXML oxide